

Hey, everybody. Welcome to your chapter 7 review. This is one of our most critical chapters. Why? In order to achieve improved physical fitness, you have to move. So fitness professionals need to have at least a fundamental understanding of the science of human movement and how it can be used and manipulated to help clients reach their goals.

The Human Movement System, or HMS, is an integrated functioning of the nervous system, skeletal system, and the muscular system, designed to enable an individual to move throughout their daily life.

Although separate in structure and function, the HMS relies on a collaborative effort to create interdependent links that form an efficient kinetic chain. Think about how your arm, shoulder, and spine are interconnected segments that function together to perform movement.

If any part of the kinetic chain is injured or not working properly, a problem may arise. In this case, the entire link is compromised, resulting in less-than-optimal performance. When moving from one body segment to the next, their movements must be coordinated to allow for efficient transfer of energy and power throughout the body.

Learning new movements requires the cooperative efforts of all systems in the HMS, appropriate feedback mechanisms, and the ability to identify abnormal movement patterns. To correct, develop, or advance movement patterns, you need to understand feedback mechanisms as well as have a working knowledge of biomechanical principles used to augment traditional exercises.

By the end of this chapter, you should be able to summarize the scientific terminology that governs exercise and movement, identify key biomechanical concepts that define how the neuromuscular system functions, as well as describe motor behavior and how the neuromuscular system adapts to changes in the environment.

We've only just scratched the surface. Now let's get into more detail. We'll start with biomechanics. Biomechanics is the study of mechanical laws or principles relating to movement, specifically how the body creates movement at the joints during exercise and physical activity.

It's important to keep in mind that biomechanical terms always refer to the body in the anatomic position, or when a person is standing with their arms hanging down by their sides and palms facing forward. This gives us a universal baseline for all of us in the allied health industry to accurately discuss human biomechanics.

It's also important to know how the body moves in three dimensions, based on a system of three imaginary planes of motion known as sagittal, frontal, and transverse planes. It's critical to understand these planes in order to accurately select exercises and design programs that meet the needs of your clientele.

The sagittal plane is an imaginary line that bisects the body into right and left sides. Movements in the sagittal plane include flexion and extension and plantar flexion and dorsiflexion. Examples of exercises include Bicep Curls, Tricep Pushdowns, Squats, Front Lunges, and more.

The frontal plane bisects the body to create front and back halves. Movements in the frontal plane include abduction and adduction of the limbs, lateral flexion of the spine, and eversion and inversion at the foot and ankle complex. A few examples of frontal plane exercises are Jumping Jacks, Side Lunges, Lateral Shoulder Raises, and Side Shuffling.

Finally, the transverse plane bisects the body to create upper and lower halves. Movements in the transverse plane include internal rotation and external rotation of the limbs, right and left rotation of the head and trunk, horizontal abduction and horizontal adduction of the limbs, and radioulnar pronation and supination.

Cable Trunk Rotations, Dumbbell Chest Flies, and swinging a bat or golf club are all examples of transverse plane exercises.

Movement in each plane occurs on an axis running perpendicular to that plane, a lot like the axle that a car wheel revolves around. We tend to simply call it joint motions, termed for their action in each of the three planes of motion.

More specific terms used to describe motion are osteokinematics, the movements we see, and arthrokinematics, the movements taking place inside the joint itself that we can't see. "Osteo" means relating to the bones. "Arthro" means relating to joints.

For example, the motions of the scapulae are important to be familiar with to help ensure proper movement of the shoulder complex during exercise. Isolated motions of the scapulae include scapular retraction, when the blades come close together, scapular protraction, when the blades move further away from each other, scapular depression, when the blades move downward, and scapular elevation, when the shoulder blades move upward toward the ears.

Be sure you understand what each term means and try to think of exercises that demonstrate each one. Movement can be described in terms of the predominant plane and the specific motion of the joints. However, movement can also be classified as normal or dysfunctional.

Normal movement represents efficient, cooperative functioning of the skeletal, nervous, and muscular systems, where joints move through their full range of motion. There are three overarching types of muscle actions.

Isotonic muscle action can be described as the movement occurring through a given range of motion when force is produced and muscle tension is developed. Isotonic muscle actions can be subdivided into either concentric or eccentric muscle action, depending on if a source of resistance is moving with or against gravity.

Isometric muscle tension is created without a change in muscle length and no visible movement of the joint. Anytime the body is in motion, numerous muscles take turns creating isometric actions to dynamically stabilize the peripheral joints and spine.

Lastly, isokinetic muscle action is when the speed of movement is fixed, and resistance varies with the force exerted. This requires sophisticated training equipment, often seen in rehabilitation clinics or exercise labs.

Since almost every exercise consists of an eccentric, isometric, and concentric muscle action to complete one repetition, the movement through those muscle actions is referred to as the muscle action spectrum.

Muscles can also play the role of agonist, synergist, stabilizer, or antagonist, depending on the functional movement and joint motion performed. Agonists are muscles that act as prime movers for a joint motion, like the quadriceps for knee extension.

Synergist muscles create forces to assist prime movers, such as the serratus anterior, assisting the anterior deltoid with shoulder flexion. Stabilizer muscles contract isometrically to support and stabilize the joints, while the agonists and synergists move through the muscle action. Think internal obliques or transversus abdominus during a squat.

Lastly, antagonist muscles perform the opposite action of the prime mover, like the biceps brachii to the triceps brachii during elbow extension.

In addition to describing an exercise based on the type of muscle movements, exercises are often classified based on different loading or weight-bearing characteristics. These are known as either closed-chain or open-chain movements.

Basically, closed-chain movements anchor the body to the ground or an object, whereas open-chain movement involves the distal limb moving freely in space. Closed-chain exercises are more likely to recruit multiple muscle groups and joints. And open-chain movements are more likely to isolate a particular joint or muscle.

A Pull-Up is an example of a closed-chain exercise, whereas the Lat Pull Down is an open-chain exercise for the same muscle group.

Muscles are influenced by outside forces from a multitude of directions. To compensate, muscles produce corresponding forces in groups to move bones and joints. The synergistic action of these muscles is known as a force couple relationship. However, the amount of force that can be produced depends on leverage.

Leverage directly affects rotary motion and torque. Modifying leverage can actually be a means of exercise progression or regression. Advanced exercises like plyometrics take advantage of the mechanical and neurological properties of muscle and tendon using the stretch shortening cycle, which increases force production and power.

Muscles also work together throughout the body, forming a network of interworking systems that both stabilize and create movement in all three planes of motion. The two overarching categories are local and global muscular systems.

The local muscular system involves muscles that attach to the spine and provide stability for the LPHC. The global muscle system can be broken down into subsystems which include the deep longitudinal, posterior oblique, anterior oblique, and lateral subsystems.

The subsystems describe the integrated function of muscle groups to transfer force for complex multi-joint movements and stabilization of the HMS. Subsystems work synergistically to perform efficient movements and stabilize the spine and pelvis.

Keep in mind that all movement relies on the nervous and skeletal systems as well. A deficit in joint function or failure of the nervous system to supply the appropriate signal can potentially lead to improper movement patterns, even if all muscles in a subsystem have ideal length tension relationships.

The amount of force produced by the HMS relies not only on muscle recruitment but also the lever type of the joint that is moving. Lever systems are classified as first-, second-, and third-class, depending on the relations among the fulcrum, the effort, and the resistance.

Third-class levers are the most predominant levers in the human body. As a fitness professional, it's important to understand that placing a muscle in a shortened position or lengthening a muscle beyond optimal length can reduce force output because optimal length is the position with maximal overlap of actin and myosin filaments.

Understanding length tension, force couple relationships, and their influence on and by posture is critical to how a client performs, experiences, and responds to exercise. When it comes to learning and your approach to how your clients learn, keep in mind there is a neurological and physical process that the HMS goes through.

Each system is interdependent, and all of the segments and processes must work together to gather information, to create and learn movements. The body uses proprioception, which is the intrinsic awareness of movement and bodily position in space, as well as sensory motor integration and muscle synergies to create efficient movement.

Then, with repeated practice and internal and external feedback, efficient movement can be reproduced. When we talk about muscle synergies, it means the cooperative function of multiple muscles recruited by the nervous system to complete any given movement pattern.

And motor learning is really just the integration of motor control processes with practice and experience leading to a relatively permanent change in the capacity to produce skilled movements. It's essentially movement memory. Feedback can come from internal or external sources and aid the process of motor learning.

Ultimately, the science of human movement highlights the philosophy that everything is connected and that the body should not be viewed in isolation. Once you know what you're looking for, it's relatively easy to recognize the regional interdependence model, that concept of how the functioning of one segment of the body can impact other areas.

Learning functional anatomy and the terminology associated with different locations and movements within the body is critical. And don't skimp on your study of the muscular subsystems, which are often overlooked, and whose role in relation to exercise programming is just as important as the global view.

And finally, remember to focus on fundamental biomechanical principles such as lever length, as well as on the process of motor learning. These will directly influence the ease or difficulty of particular movements and exercises, how you select and design exercise programs, and how you work with your clients to coach, help them progress, or even take a step back if needed.

Right now, you've just taken a big step forward in your knowledge and expertise. There is so much to learn and explore about movement. And every bit of that energy you put toward fitting the pieces together will be what makes you a next-level fitness pro. So keep going, and happy studying.